School of Engineering

College of Arts and Sciences

WiSeMAN Research Lab Organizes

Fall 2026 International Distinguished Lecture Series

Understanding Malicious Code and Vulnerabilities in Scale and Depth

Distinguished Speaker: Dr. Constantinos Patsakis

Professor of Informatics

University of Piraeus, Piraeus, Greece

Date: August 27, 2026

Time: 12:00 pm – 1:00 pm

Venue: Online

Abstract: The rapid proliferation and increasing sophistication of modern malware pose fundamental challenges to both automated defense mechanisms and threat analysis. Effective mitigation requires examining malicious software across two complementary dimensions: at scale, evaluating how automated machine learning (ML) systems detect and classify malicious binaries in large corpora, and in depth, analyzing malware authors' source code, intrinsic vulnerabilities, and development strategies. At scale, static ML-based detectors, spanning handcrafted feature models (e.g., EMBER LightGBM), end-to-end deep learning (MalConv), and computer vision architectures (ResNet, SwinTransformer), frequently exhibit severe performance degradation when exposed to binary packing and obfuscation. Structural analyses of benchmark datasets (such as Malimg and BODMAS) demonstrate that classifiers often learn packer-specific artifacts rather than underlying malicious functionality. This reliance creates substantial evasion vulnerabilities and elevates false-alarm rates when encountering benign packed software or novel runtime protectors. In depth, static application security testing (SAST) and software engineering metrics applied to leaked malware codebases reveal the cognitive and operational realities of malware development. Compared to benign open-source software, malware codebases exhibit minimal documentation, low modularity, higher per-function cyclomatic complexity, and a distinct prevalence of structural weaknesses (such as CWE-398, uninitialized variables, and missing integrity checks). Furthermore, malware authors regularly introduce severe vulnerabilities within their own code, including cleartext command-and-control channels and disabled certificate verification, reflecting development strategies optimized for expedience, evasion, and disposable lifecycles under operational secrecy and time constraints. By synthesizing macro-level static detection limits with micro-level source code and vulnerability analysis, this presentation outlines the structural blind spots in ML pipelines and presents strategic opportunities for defensive disruption. Grounded in empirical evaluations across hundreds of malware families and multiple packers, the talk demonstrates how combining binary robustness testing with behavioral software analysis provides a comprehensive framework for proactive malware defense. Finally, it will present how malware sandboxes can be exploited to bypass their telemetry mechanisms and report malware as benign software.

Speaker's Biography: Constantinos Patsakis received a B.Sc. in mathematics from the University of Athens, Greece, an M.Sc. in information security from Royal Holloway, University of London, and a Ph.D. in cryptography and malware from the University of Piraeus. In the past, he has worked as a researcher at the UNESCO Chair in Data Privacy, at Rovira i Virgili, at Trinity College, Dublin, and the Luxembourg Institute of Science and Technology. He is a Professor at the University of Piraeus and an Adjunct Researcher at the Athena Research and Innovation Center. He is a senior IEEE member. He has authored more than 150 publications in prestigious peer-reviewed international conferences and journals and participated in several national and European Research and Development projects. His primary areas of research include cryptography, security, privacy, blockchain technology, and cybercrime.



All Faculty, Staff, and Students are most welcome to attend this exciting talk!

For more information about this Fall 2026 International Distinguished Lecture Series, please contact Prof. Habib M. Ammari (Endowed Professor of Computer Engineering, Founding Coordinator) at Habib.Ammari@tamiu.edu